BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 142 · Lesson 142 · Networks: LANs, WANs, Standards and Protocols

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which ACID property keeps committed transaction changes after a crash?

Q2. How many bits are in one byte?

Q3. What data structure uses LIFO?

Q4. What is a trace table used for?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LAN — Network spanning a relatively small geographical area.

WAN — Network connecting networks over larger distances.

protocol — Agreed rules for communicating between systems.

standard — Shared specification that enables interoperability.

New learning

Identify scope — LAN connects local devices; WAN connects distant sites.

Interconnect devices — Switches connect local devices; routers forward traffic between networks.

Use open rules — Protocols and standards let different equipment communicate.

Worked example

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

1. Same school buildings — Identify the relevant information and make a prediction.

2. Local switched/Wi-Fi network — Explain the change using correct technical terms.

3. Standard protocols — A school LAN connects local devices. Ethernet/Wi-Fi standards support compatible links and IP-based protocols allow data exchange. A router becomes relevant for communication beyond the local IP network.

We Do · Practise with guidance

A trust connects schools 40 miles apart using provider links. Classify the network.

Separate school LANs

Identify the relevant information and make a prediction.

Router/provider link

Explain the change using correct technical terms.

Connected WAN

It is a WAN connecting multiple site LANs. Routers and service-provider infrastructure carry traffic between sites; secure connections and availability matter.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Differentiate LAN and WAN with examples. [6]

Question 2

Explain why protocols and standards matter when devices use different manufacturers. [6]

Question 3

A school requires a shared printer, wireless access and broadband internet. Explain suitable network devices and their functions. [8]

Exit Ticket · Answer before checking

A WAN is defined by having more devices than a LAN.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Durability.

Q2. Eight.

Q3. Stack.

Q4. Tracking variable state changes as an algorithm runs.

You Do pages · Independent questions

Q1. LAN limited local scope [2] and example [1], WAN wider link of networks [2] and example [1].

Q2. Common data formats/behaviours [2], compatibility/interoperability [2], reliable predictable exchange [2].

Q3. Switch [2], wireless access point [2], router [2], justification of links/traffic [2].

Exit Ticket · Model answer

False. The distinction is geographical scope/connection of distant networks.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: